
GRAFIKA KOMPUTEROWA - LAB 12

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Joanna Dagil 231008

```
clear; close all; clc;
```

CEL CWICZENIA

Transformata Fouriera i szybka transformata Fouriera (FFT).

ZADANIA

```
% -----
```

ZADANIE 1

```
-----
```

```
for n = 0:1023
    x(n+1) = cos(0.314*n + 2.5) + 2*cos(1.57*n - 0.18);
end

xx(1:32) = x(1:32);           % krotka sekwencja
w = 0:0.001:2*pi;           % "ciagly" zakres czestotliwosci
[aux,k] = size(w);

for i = 1:k
    ww(i) = 0 + 1*0;
    for j = 0:31
        ww(i) = ww(i) + xx(j+1)*exp(-1*j*w(i)); % ze wzoru slajd 11
    end
end

figure(1);
plot(w, abs(ww))
hold on;
```

```
fww = fft(xx); % FFT
for i = 1:32
    p1(i) = (i-1)*2*pi/32;
end

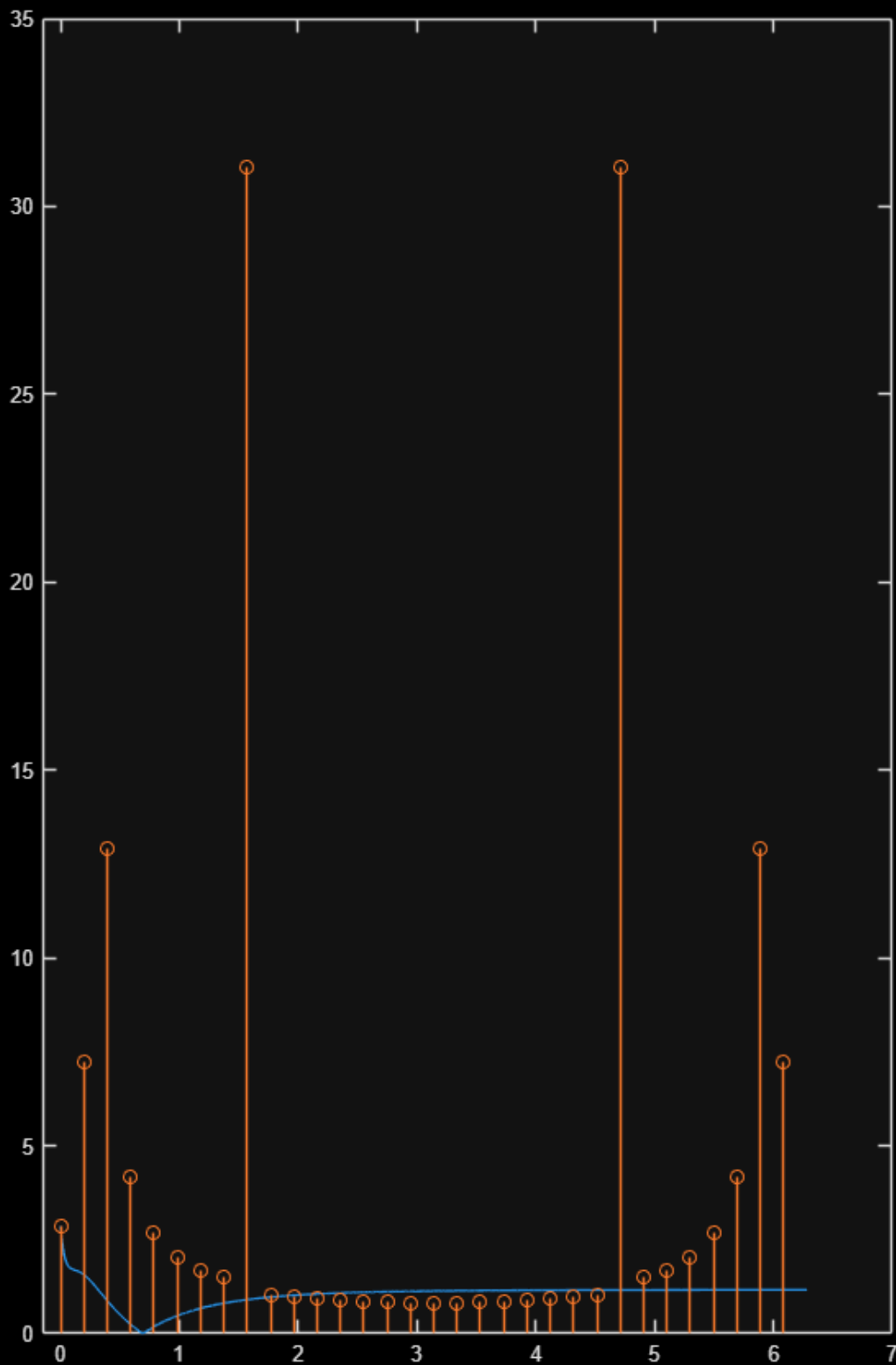
stem(p1, abs(fww));
hold off;

figure(2);
plot(p1, abs(fww));

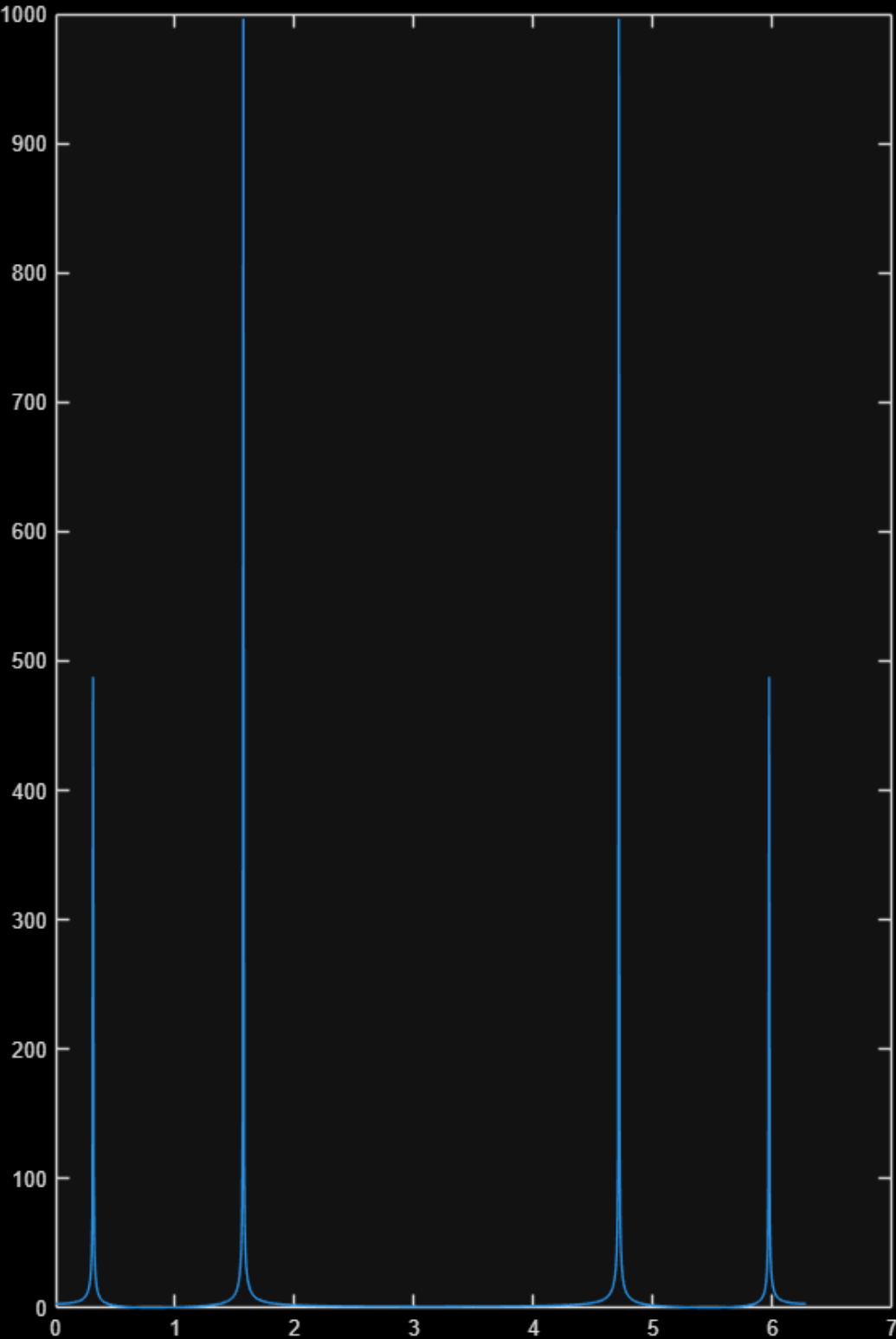
% 1024 probki
fw = fft(x);
for i = 1:1024
    p12(i) = (i-1)*2*pi/1024;
end

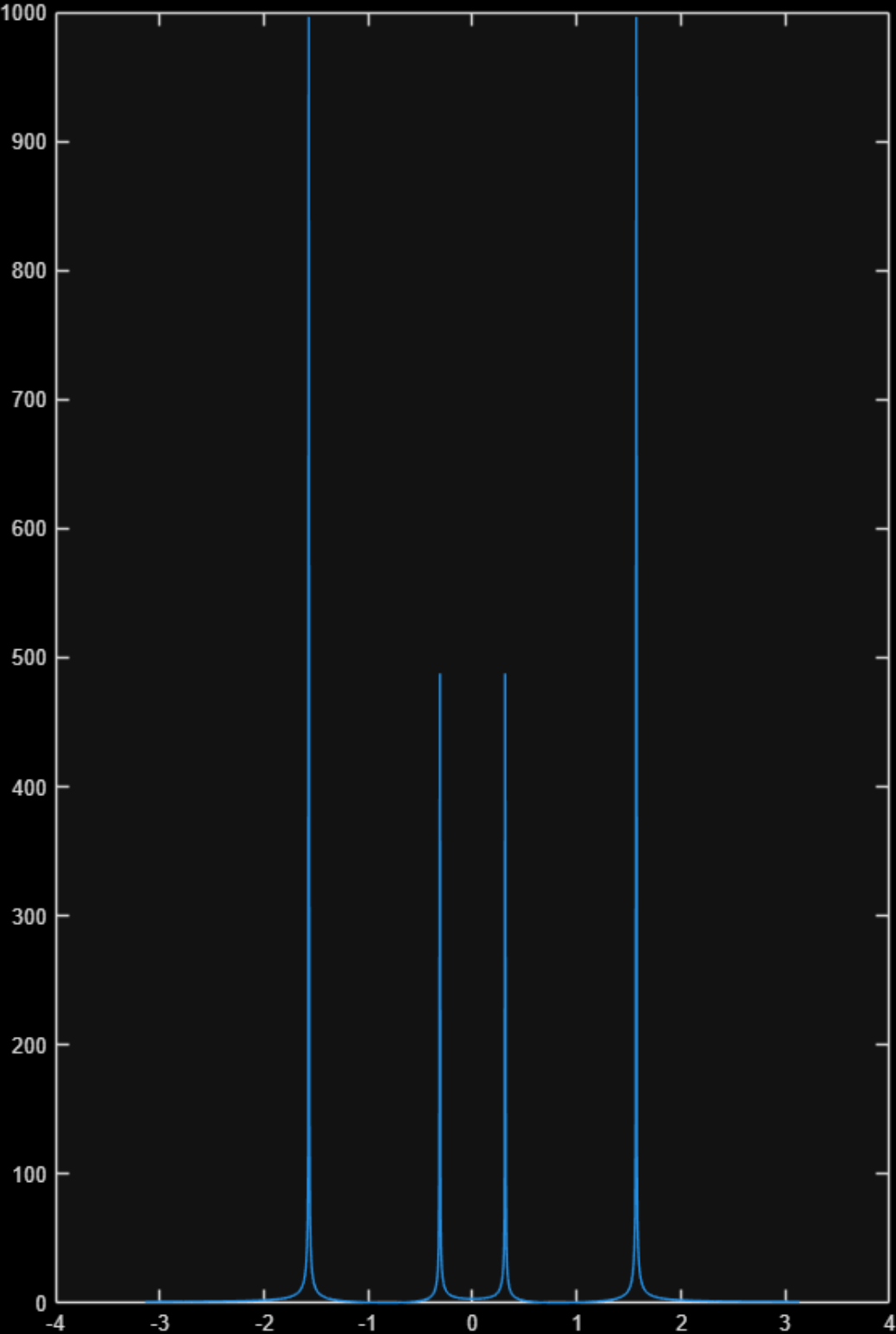
figure(3);
plot(p12, abs(fw));
fwsh = fftshift(fw);
figure(4);
plot(p12-pi, abs(fwsh));
```

```
% -----
```









ZADANIE 2

```
[music, f] = audioread('icing2.wav');

[K, aux] = size(music);
fmusic = fft(music);
figure(5); plot(abs(fmusic));

% zamiana wsp. x na herce
for i = 1:K
    hz(i) = f*(i-1)/K;
end

hz2 = hz - 22050;
fmusic2 = fftshift(fmusic);
figure(6); plot(hz2, abs(fmusic2));

fmusic2_low = fmusic2;

for i = 1:K
    if abs(hz2(i)) > 2205
        fmusic2_low(i) = 0;
    end
end

figure(7); plot(hz2, abs(fmusic2_low));

fmusic_low = ifftshift(fmusic2_low);
music_low = ifft(fmusic_low);

audiowrite('icing2low.wav', music_low, f);

fmusic2_high = fmusic2;

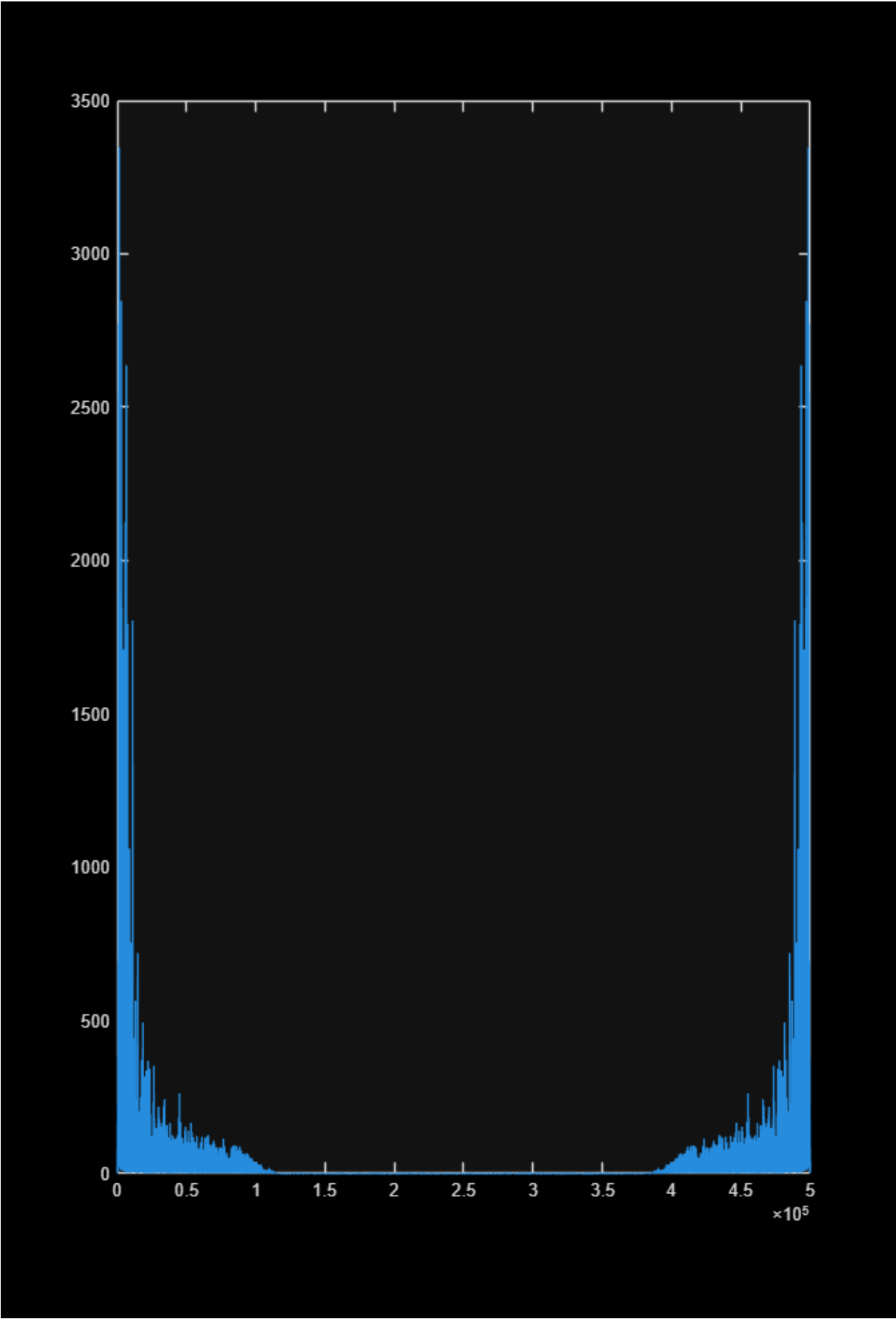
for i = 1:K
    if abs(hz2(i)) < 1103
        fmusic2_high(i) = 0;
    end
end

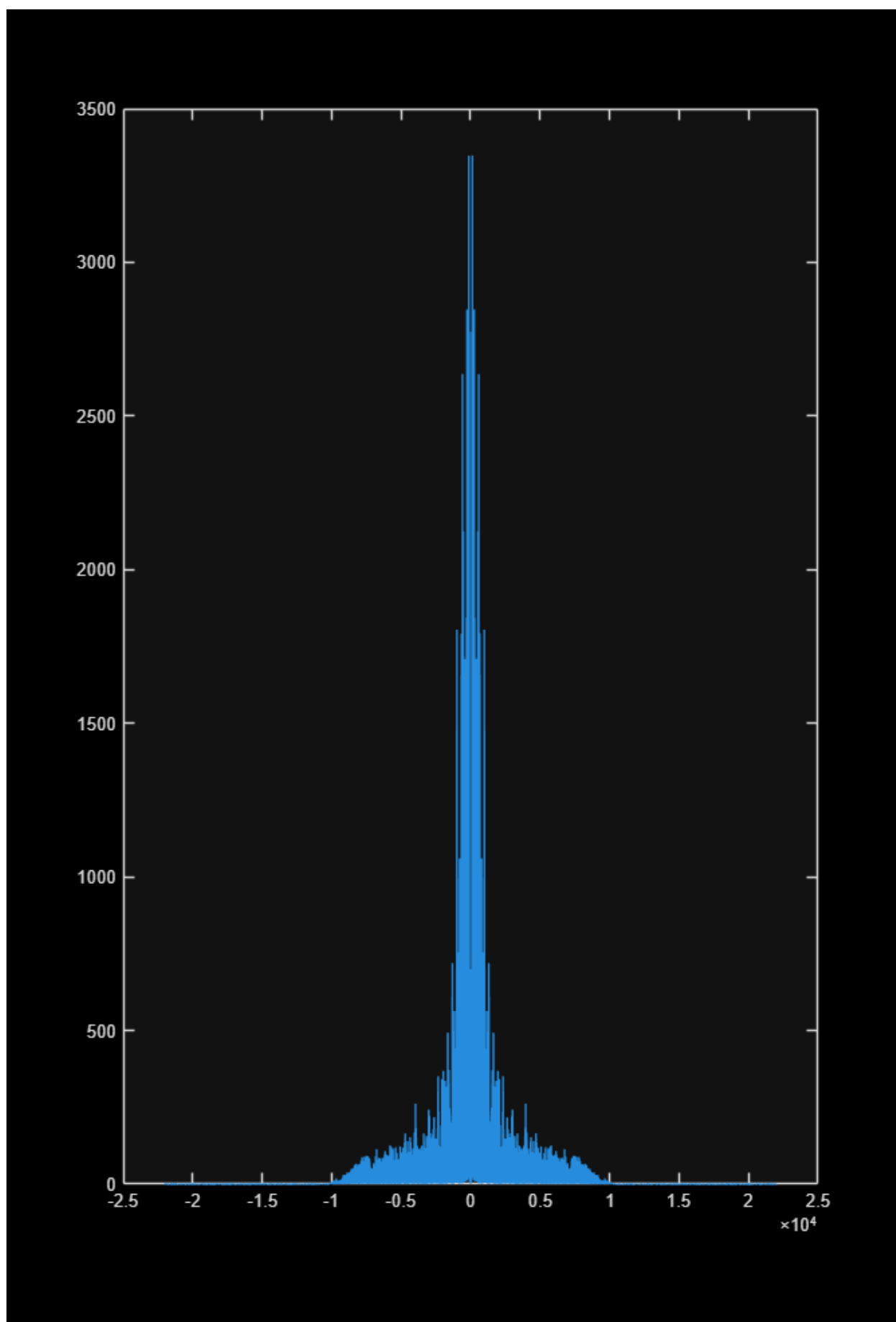
figure(8); plot(hz2, abs(fmusic2_high));

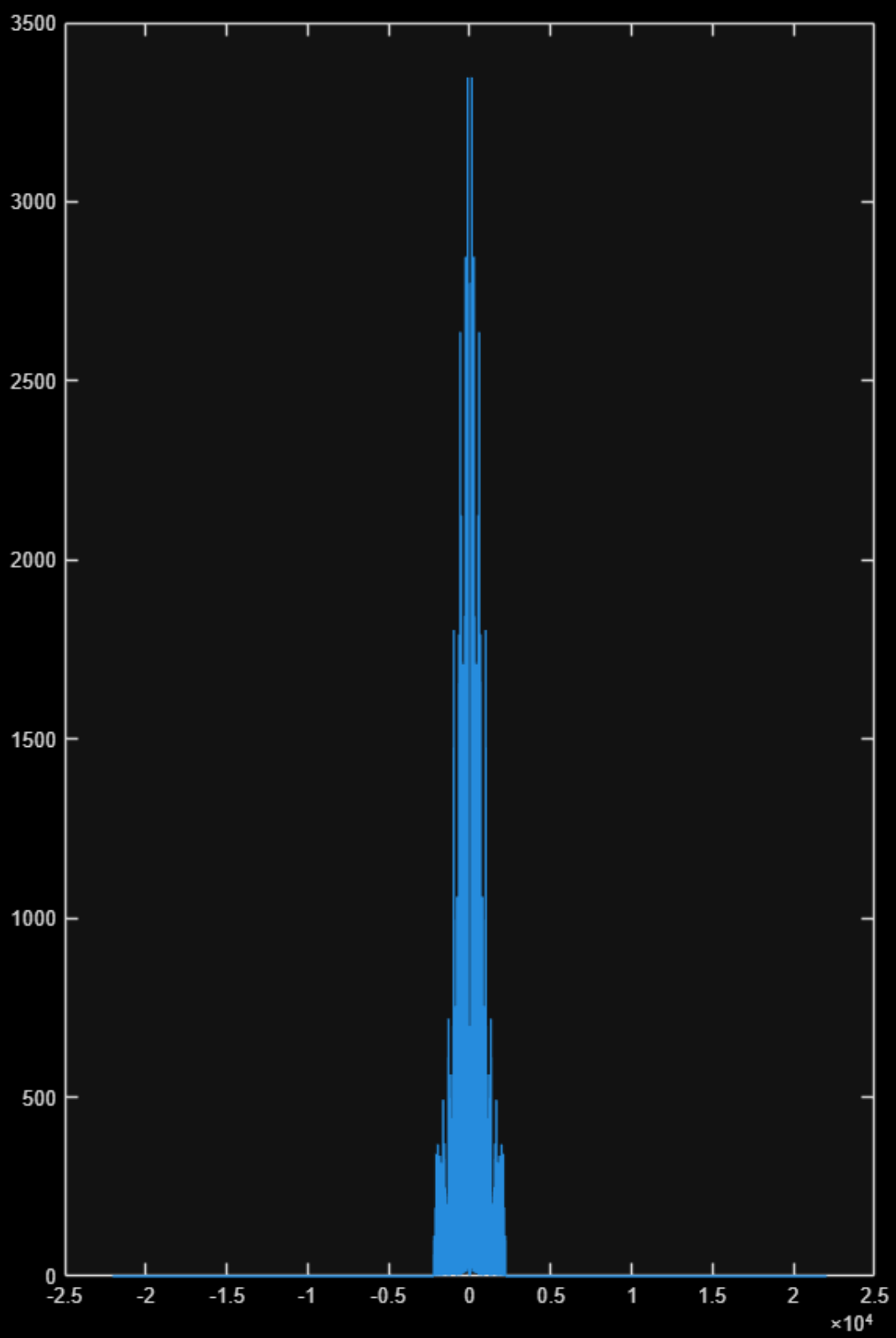
fmusic_high = ifftshift(fmusic2_high);
music_high = ifft(fmusic_high);

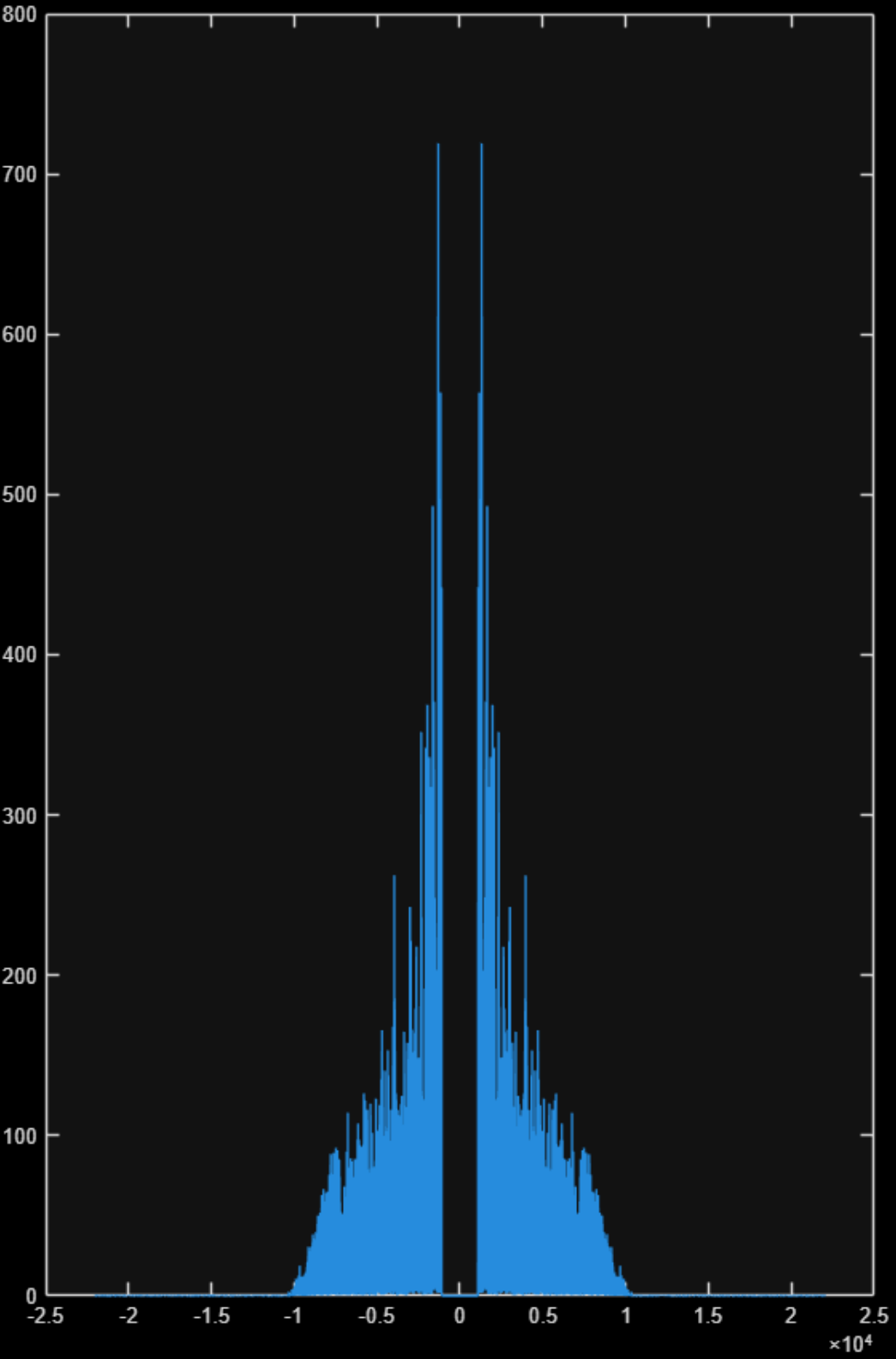
audiowrite('icing2high.wav', music_high, f);

% -----
```





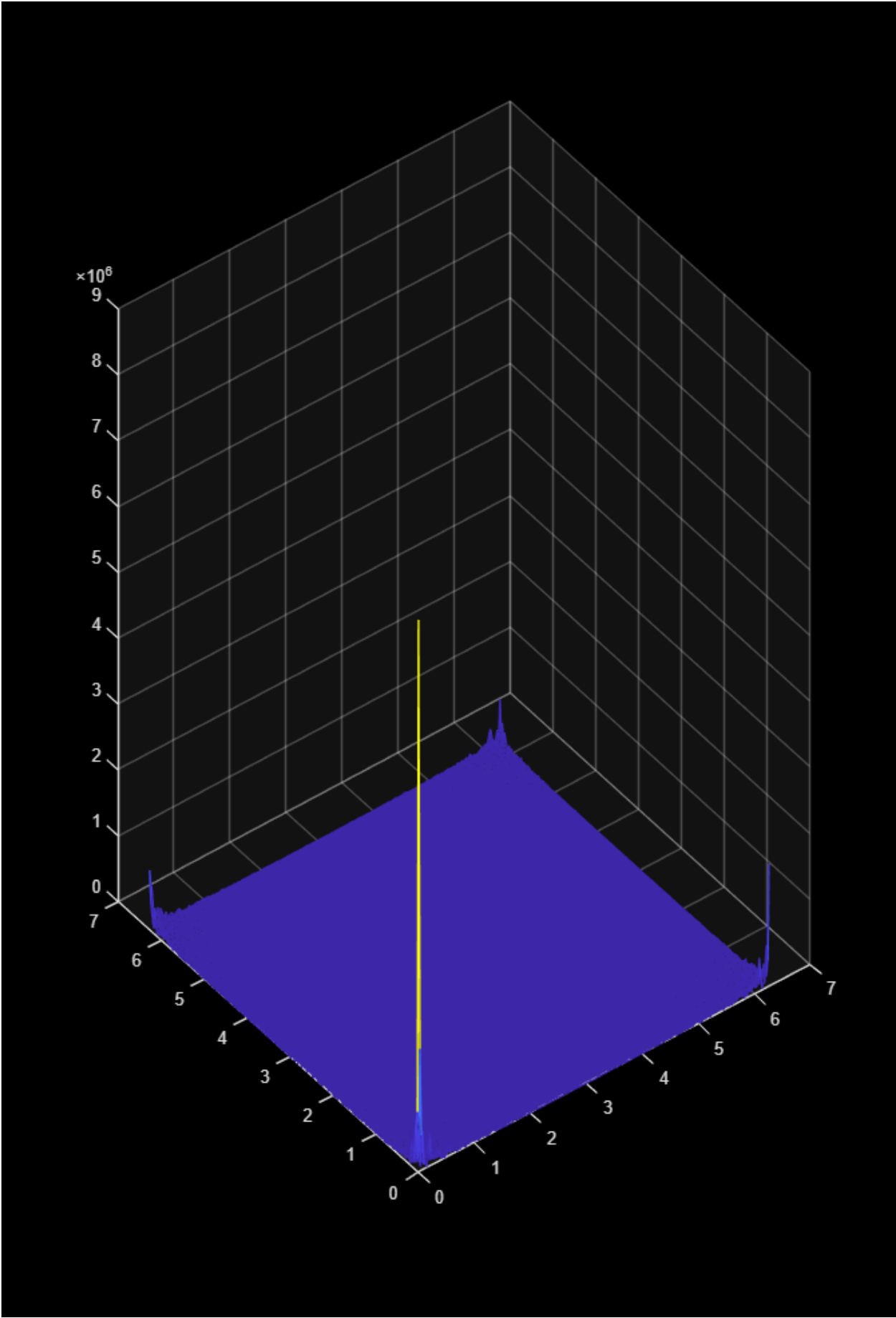


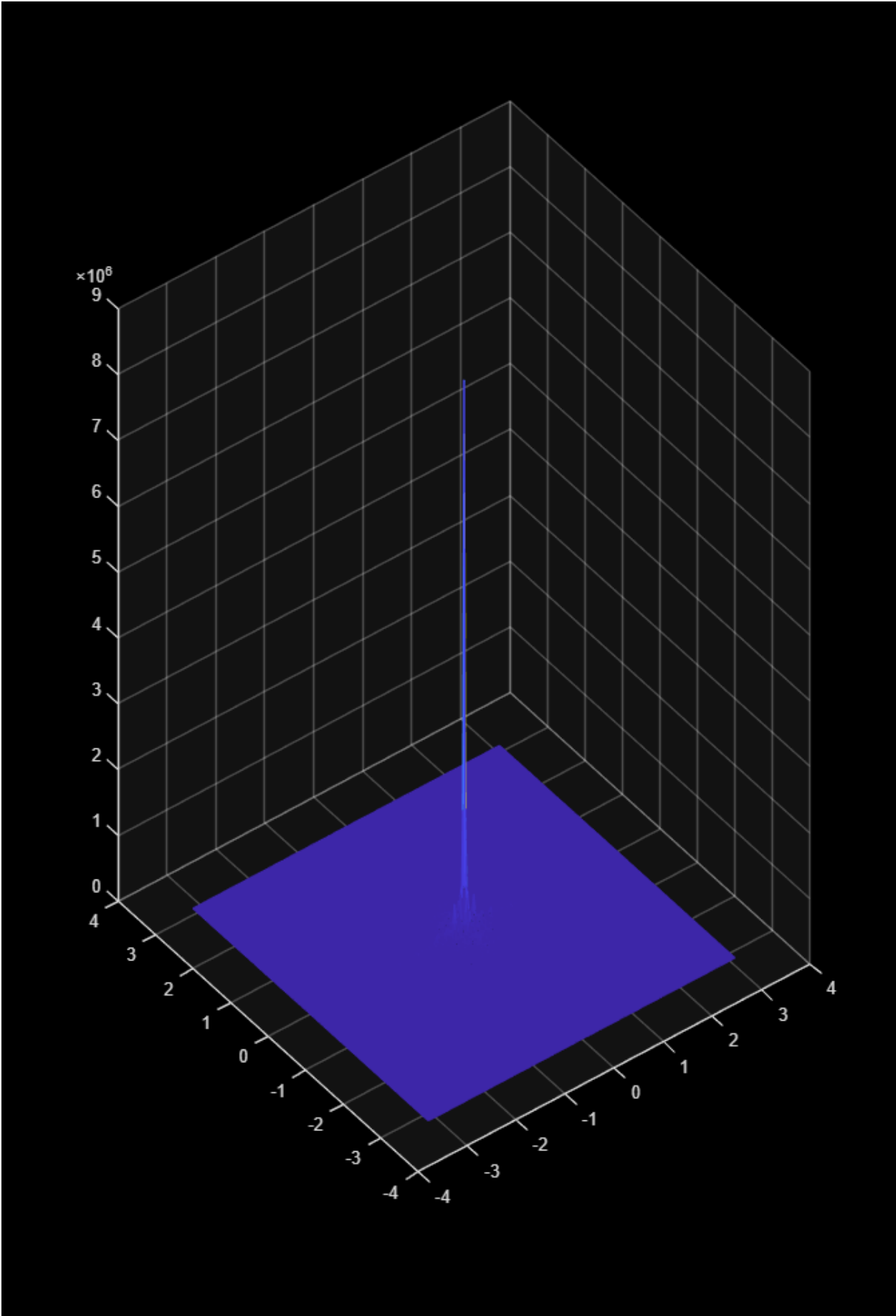


ZADANIE 3

```
-----  
Im = imread('LAKE.bmp');  
figure(9); imshow(Im);  
dIm = double(Im);  
fIm = fft2(dIm);  
% zakres czestotliwosci od 0 do 2PI  
for i = 0:255  
    wX(i+1) = i*2*pi/256; % rozdzielczosc obrazka 256x256  
    wY(i+1) = i*2*pi/256;  
    wXsh(i+1) = i*2*pi/256-pi; % czestotliwosci dodatnie i ujemne  
    wYsh(i+1) = i*2*pi/256-pi; % czestotliwosci dodatnie i ujemne  
end  
fImsh = fftshift(fIm);  
figure(10); mesh(wX,wY,abs(fIm));  
figure(11); mesh(wXsh,wYsh,abs(fImsh));  
  
% -----
```







ZADANIE 4

```

Im = imread('LAKE.bmp');
dIm = double(Im);
fIm = fft2(dIm);
% zakres czestotliwosci od 0 do 2PI
for i = 0:255
    wX(i+1) = i*2*pi/256; % rozdzielczosc obrazka 256x256
    wY(i+1) = i*2*pi/256;
    wXsh(i+1) = i*2*pi/256-pi; % czestotliwosci dodatnie i ujemne
    wYsh(i+1) = i*2*pi/256-pi; % czestotliwosci dodatnie i ujemne
end
fImsh = fftshift(fIm);
figure(12); mesh(wXsh,wYsh,abs(fImsh));

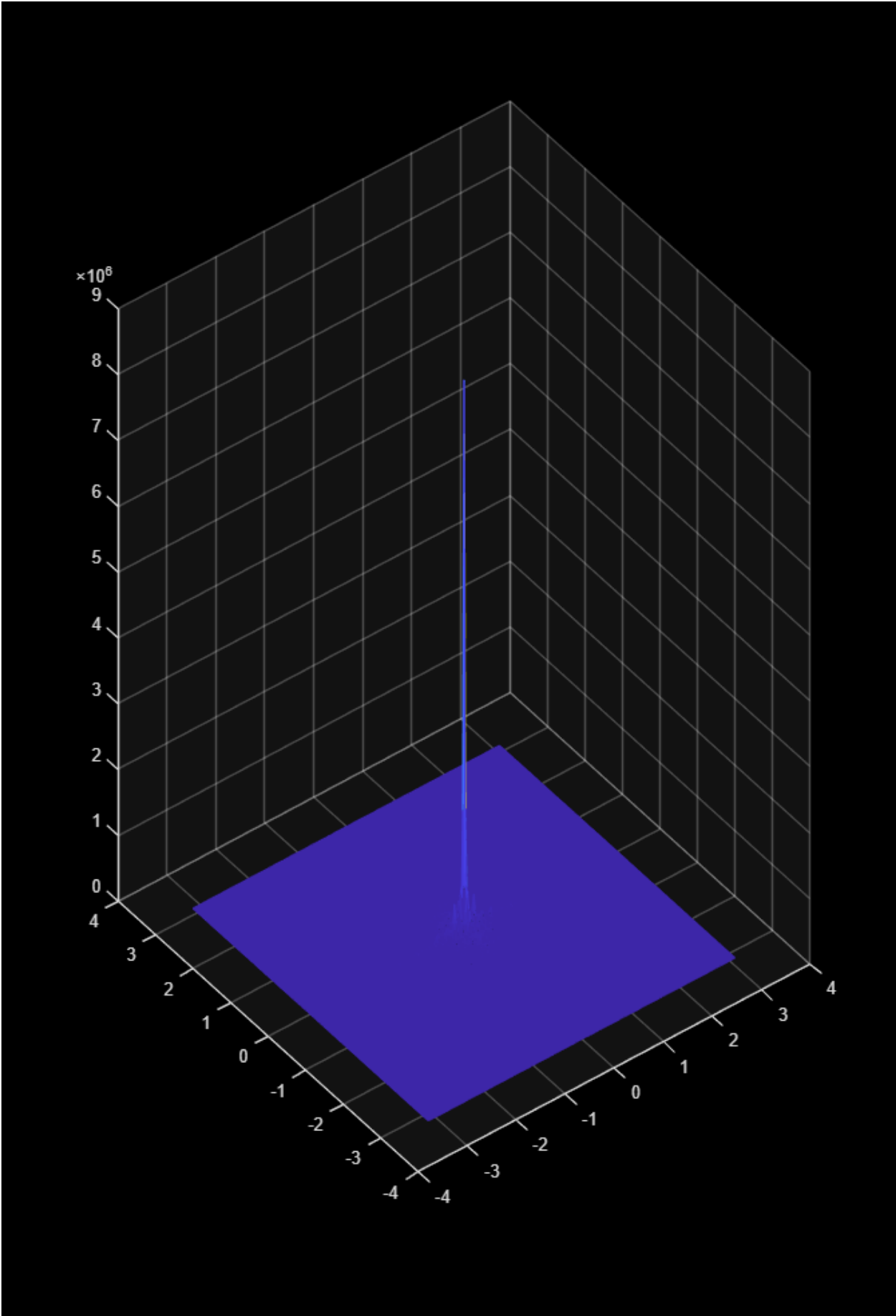
% czestotliwosc wx
fImshx = fImsh;
for n = 1:256
    if abs(wXsh(n)) > pi/12
        for m = 1:256
            fImshx(m,n) = 0;
        end
    end
end
figure(13); mesh(wXsh,wYsh,abs(fImshx));
fIm2 = ifftshift(fImshx);
Im2 = uint8(ifft2(fIm2));
figure(14); imshow(Im2);

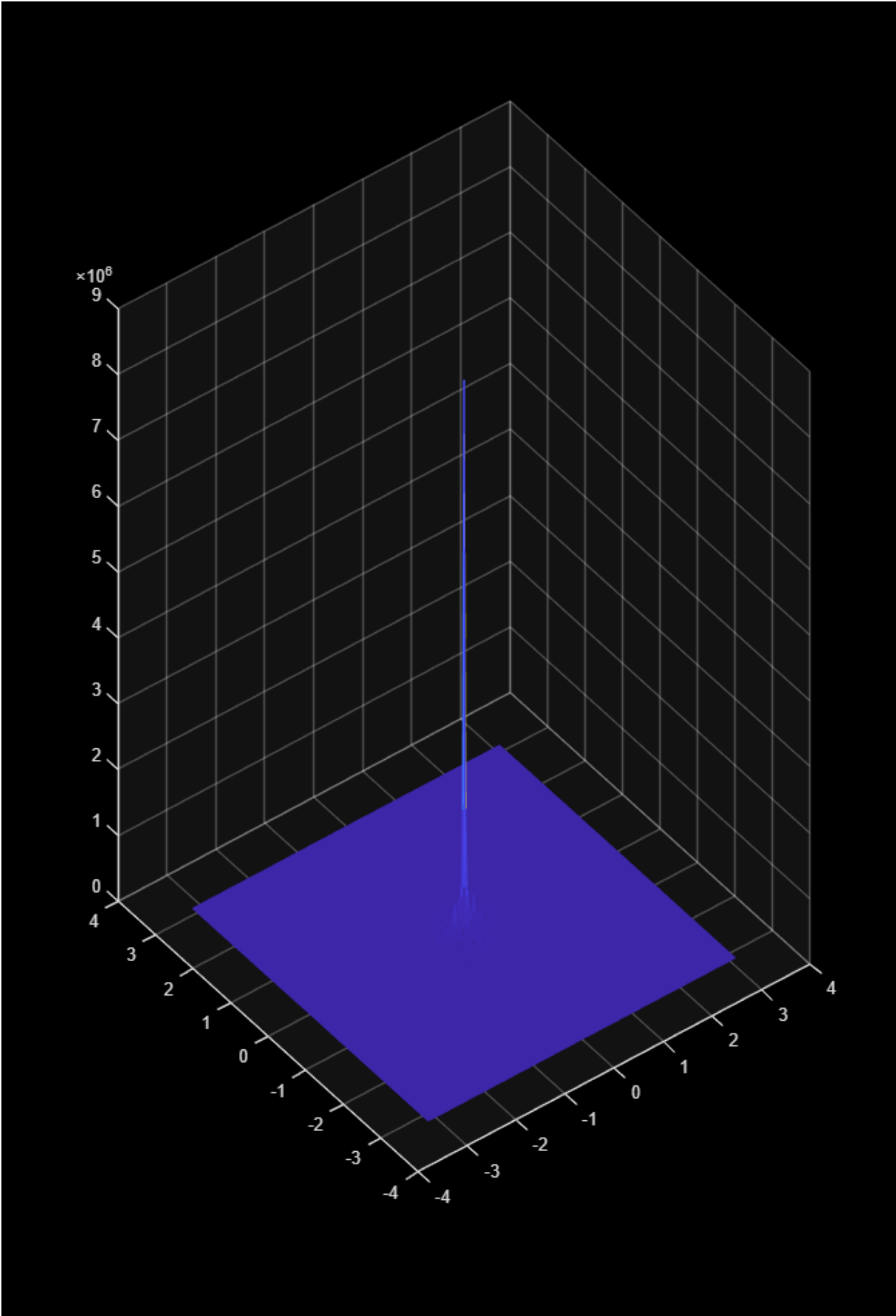
% czestotliwosci wy
fImshy = fImsh;
for n = 1:256
    if abs(wYsh(n)) < pi/12
        for m = 1:256
            fImshy(n,m) = 0;
        end
    end
end
figure(15); mesh(wXsh,wYsh,abs(fImshy));
fIm3 = ifftshift(fImshy);
Im3 = uint8(ifft2(fIm3));
figure(16); imshow(Im3);

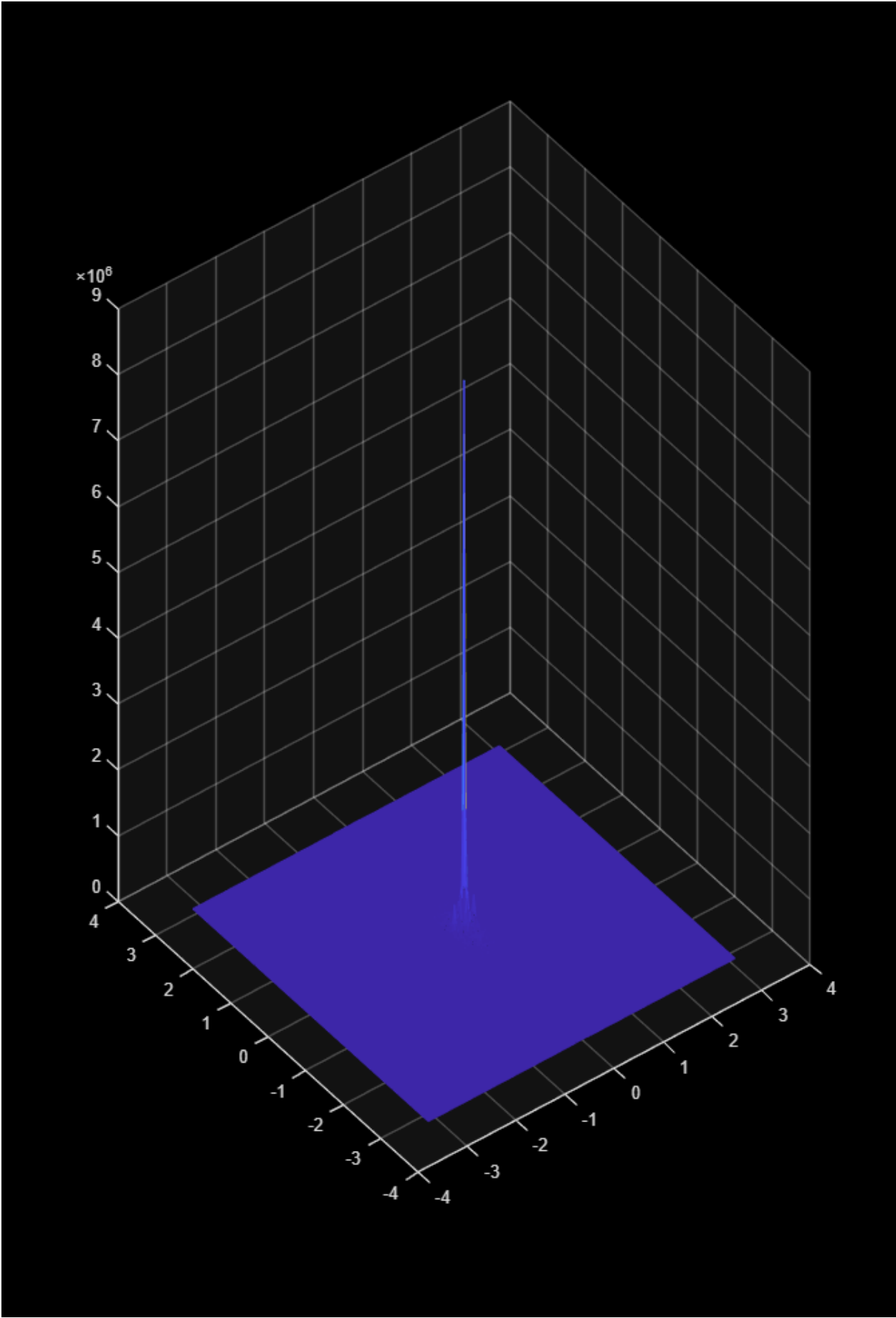
% czestotliwosci wx i wy
fImshxy = fImsh;
for n = 1:256
    for m = 1:256
        if abs(wYsh(n)) > pi/10 || abs(wXsh(m)) > pi/10
            fImshxy(n,m) = 0;
        end
    end
end

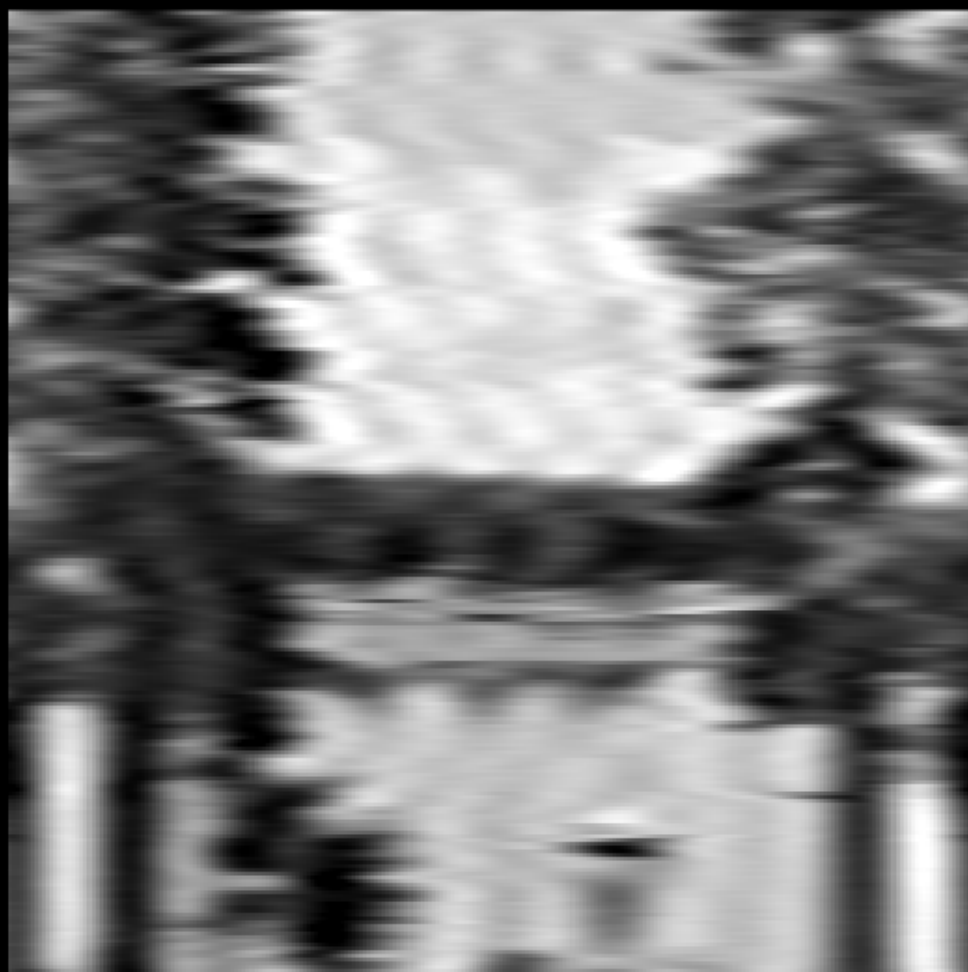
```

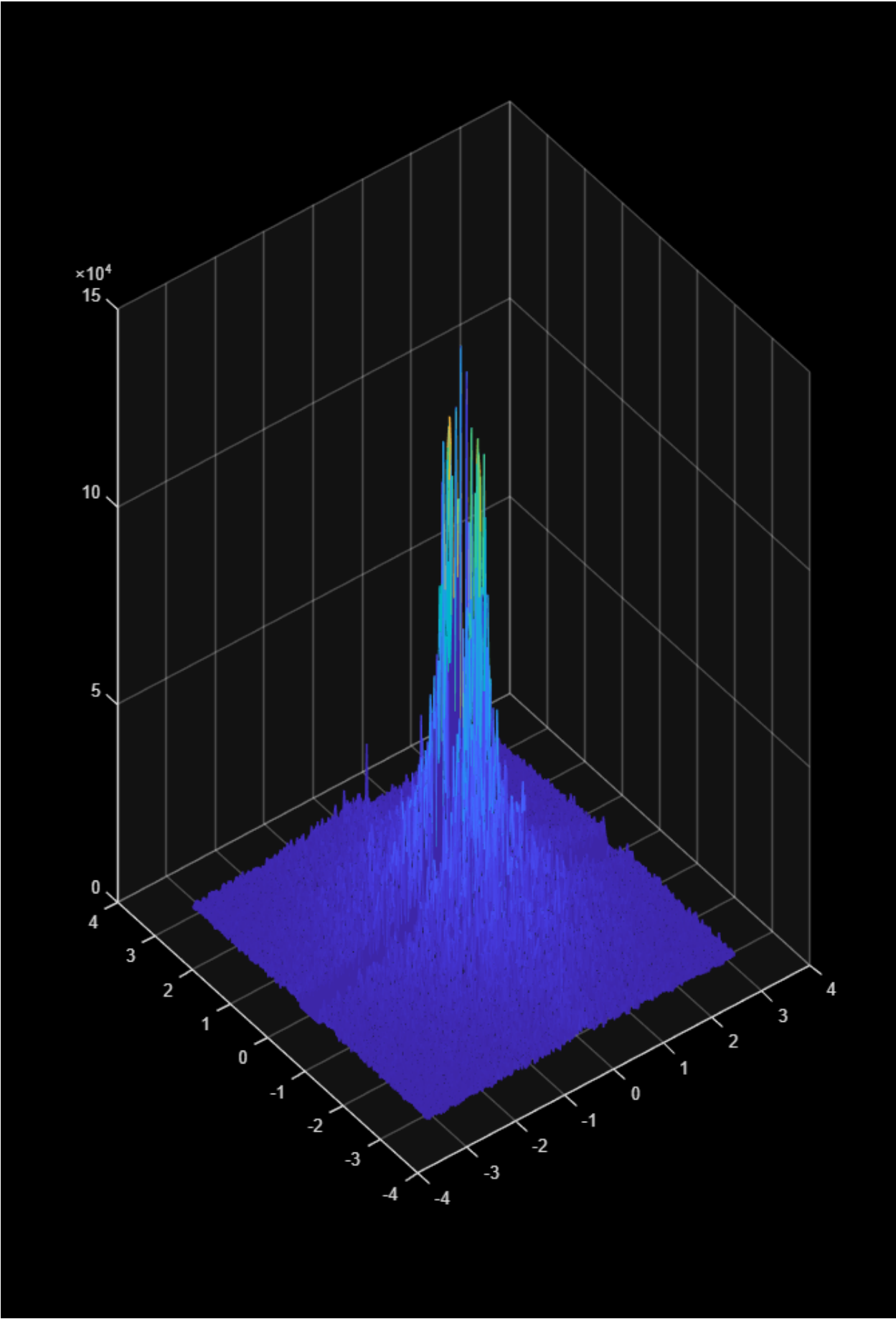
```
end  
figure(17); mesh(wXsh,wYsh,abs(fImshxy));  
fIm4 = ifftshift(fImshxy);  
Im4 = uint8(ifft2(fIm4));  
figure(18); imshow(Im4);
```



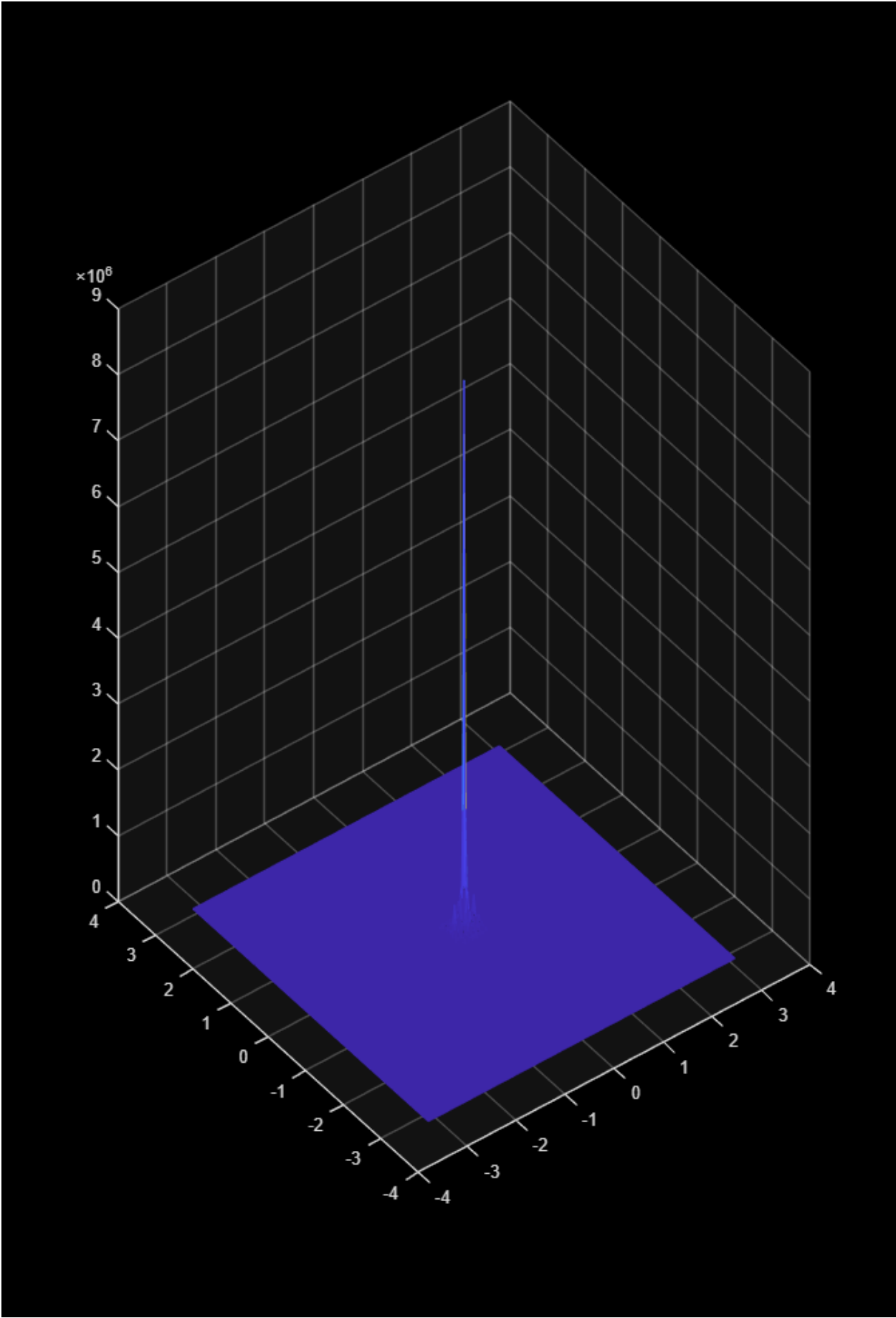














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